Applications of Higher Order Partial Derivatives

Trim Ch 11, S. 12.5

https://www.thestrad.com/video/violin-string-vibrations-captured-on-video/5897.article

Outline:

1. The Laplace Equation
2. The Wave Equation

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The Laplace Equation

The three-dimensional Laplace Equation is given by,

$$\frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} + \frac{\partial^2 f}{\partial z^2} = 0$$

The Laplace Equation can also be used in two-dimensions, which is given as,

$$\frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} = 0$$

Solutions of this equation are called **harmonic functions**.

Harmonic functions play a role in problems of heat conduction, fluid flow and electrostatic potential (see examples 12.12 and 12.13 (page 815), Trim Textbook).

The One-Dimensional Wave Equation

The wave equation describes, among other applications, the vibration of a violin string.

It is given by,

$$\frac{\partial^2 \omega}{\partial t^2} = c^2 \frac{\partial^2 \omega}{\partial x^2}$$

where,

ω: wave height
x: distance, starting at one end of string
t: time
c: velocity at which wave propagates

Note the regular pattern of peaks and valleys in an instant of time.

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Application Problem – Midterm Level

If f and g are any twice-differentiable functions of one variable, show that,

$$\omega = f(x - ct) + g(x + ct)$$

Satisfies the one-dimensional wave equation:

$$\frac{\partial^2 \omega}{\partial t^2} = c^2 \frac{\partial^2 \omega}{\partial x^2}$$

Solution:

To show that ω satisfies the one-dimensional wave equation, we need to demonstrate that the left-hand side (LHS) is equal to the right-hand side (RHS) of the wave equation, using differentiation.

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Application Problem – Midterm Level

Because we have a composition of functions, let us do the following change of variables,

Note:

The chain rule of one variable stays that if $y = f(u)$ and $u = u(t)$,

$$\frac{df(u)}{dt} = \frac{df(u)}{du} \cdot \frac{du(t)}{dt}$$

Application Problem – Midterm Level

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Summary

- Higher order partial derivatives appear in many physical problems in the form of partial differential equations (PDEs)
- Applications include,
  - The Laplace Equation (e.g. heat conduction, fluid flow, electric potential)
  - The wave equation (e.g. vibration of strings)
- The chain rule is a very useful tool to solve PDEs but could become cumbersome as the number of variable increases

Thank you